Deep Cross-Modal Image–Voice Retrieval in Remote Sensing

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***Abstract*— With the rapid progress of satellite and aircraft technologies, cross-modal remote sensing image–voice retrieval has been studied in geography recently. However, there still exist some bottlenecks: how to consider the characteristics of remote sensing data adequately and how to reduce the memory and improve the retrieval efficiency in large-scale remote sensing data. In this article, we propose a novel deep cross-modal remote sensing image–voice retrieval approach, namely, deep image–voice retrieval (DIVR), to capture more information of remote sensing data to generate hash codes with low memory and fast retrieval properties. Especially, the DIVR approach proposes inception dilated convolution module to capture multiscale contextual information of remote sensing images and voices. Moreover, in order to enhance cross-modal similarity, the deep features' similarity term is designed to make paired similar deep features as close as possible and paired dissimilar deep features as mutually far as possible. In addition, the quantization error term is designed to drive hash-like codes to approximate hash codes, which can effectively reduce the quantization error for hash codes' learning. Extensive experimental results on three remote sensing image–voice data sets show that the proposed DIVR approach can outperform other cross-modal retrieval approaches.**

***Index Terms*— Cross-modal remote sensing image–voice retrieval, deep features' similarity, deep hash codes, quantization error.**

I. Introduction

With the rapid development of satellite and aircraft technologies, the era of remote sensing big data has begun [1], [2]. As is known to all, it is very important to mine effective information in large remote sensing data. Selecting effective information based on manual work is very time consuming, and it is impossible to complete it when the data are too large [3]. Therefore, more and more researchers pay attention to large-scale remote sensing data retrieval (RSDR) [4], [5] because RSDR can automatically mine effective information and has broad application prospects in disaster relief and other fields [6], [7].

Generally speaking, RSDR approaches can be roughly grouped into unimodal RSDR approaches and multimodal RSDR approaches. Unimodal RSDR approaches [8]–[10] mainly retrieve remote sensing data that are similar to queried remote sensing data where all remote sensing data belong to the same modality. For example, Ye *et al.* [10] introduced a flexible multiple-feature hashing learning framework for large-scale remote sensing image retrieval. Multimodal RSDR approaches mainly retrieve remote sensing data that are similar to queried remote sensing data where queried remote sensing data and remote sensing data of the searching data set come from different modalities [11]. For example, Chaudhuri *et al.* [11] proposed a novel deep neural network-based architecture that is considered to learn a discriminative shared feature space for all the input modalities. Compared with unimodal RSDR approaches, multimodal RSDR approaches are more challenging due to the heterogeneity of multimodal data. With the increasing availability of multimodal remote sensing data, multimodal retrieval has attracted more and more attention in recent years.

Cross-modal remote sensing image–voice retrieval is one of the most important aspects of multimodal data retrieval. For example, Mao *et al.* [12] proposed a novel cross-modal remote sensing (RS) image–voice retrieval approach, which integrates deep features learning and multimodal learning into a unified framework for speech-to-image retrieval. The task of cross-modal remote sensing image–voice retrieval is to utilize remote sensing images (respectively, remote sensing voices) to retrieve relevant remote sensing voices (respectively, remote sensing images). Because the feature representations of voices and the feature representations of images are inconsistent, it is difficult to retrieve effective information. How to effectively achieve cross-modal image–voice retrieval is a huge challenge. To solve the abovementioned problems, some cross-modal approaches [13]–[15] have been proposed to project all samples of each modality to common feature space and then use feature representations to perform cross-modal retrieval. However, these approaches cannot capture cross-modal correlation by using shallow projects.

Recently, with the development of deep learning, some deep image–voice retrieval (DIVR) approaches [12], [16] have been developed to use deep projects to learn cross-modal correlation. However, there are still some challenges in the DIVR approaches. First, they use deep projects to learn the

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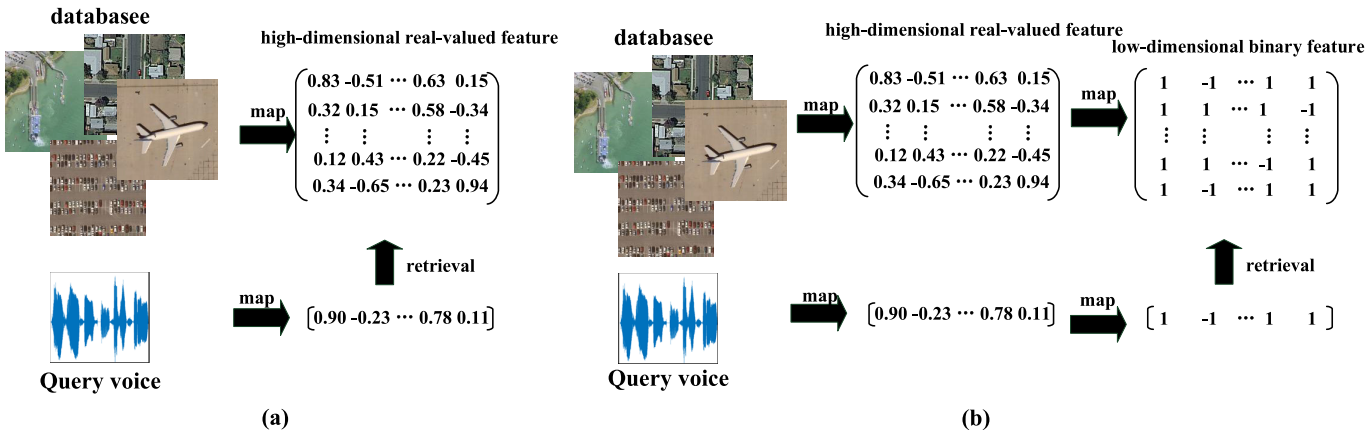


Fig. 1. Comparison of (a) deep cross-modal remote sensing image–voice retrieval approach [12] and (b) proposed DIVR approach. Mao *et al.* [12] performed cross-modal remote sensing image–voice retrieval by leveraging high-dimensional real-valued features, which needs more storage and longer retrieval time than low-dimensional binary features utilized by the proposed DIVR approach.

feature representations, and then, the feature representations are utilized to compute the cross-modal similarity. However, these feature representations are high-dimensional real-valued features that need more storage and longer retrieval time than low-dimensional binary features (i.e., hash codes). The differences between deep cross-modal remote sensing image–voice retrieval approach [12] and the proposed DIVR approach are shown in Fig. 1. Second, remote sensing images and voices often contain rich and complex structures [17]. Nevertheless, the existing method fails to consider multiscale contextual information of remote sensing images and voices adequately, which will result in the problem of insufficient utilization of data characteristics and ultimately affect the retrieval performance.

In effect, queried remote sensing images (respectively, remote sensing voices) can be more matched with relevant remote sensing voices (respectively, remote sensing images) if more information of remote sensing images (respectively, remote sensing voices) are captured by humans [18]. Clearly, the abovementioned problems would be solved if we use low-dimensional binary features to retrieve and capture more information on remote sensing images and sounds [19]. Motivated by this idea, a cross-modal remote sensing image–voice retrieval scheme is proposed to perform hash codes' learning by capturing multiscale contextual information of remote sensing images and voices.

In this article, we propose a novel deep cross-modal remote sensing image–voice retrieval approach, namely, DIVR. DIVR can generate hash codes by utilizing multiscale contextual information of remote sensing images and voices and the similarity of deep features simultaneously, as shown in Fig. 2. The whole approach contains the remote sensing image branch and the remote sensing voice branch. The remote sensing image branch contains three 2-D convolutional layers with max-pooling and batch normalization (BN), inception dilated convolution module, a fully connected layer, and a hash layer. For the remote sensing voice branch, the first convolution layer exploits filters with the width of one frame across the whole frequency axis, and then, the following layer of the remote sensing voice branch consists of three 1-D convolutions with BN and max-pooling, an inception dilated convolution module, a fully connected layer, and a hash layer. To reduce the memory and improve the retrieval efficiency, hash codes that can be learned from raw remote sensing voices and images are utilized to perform cross-modal remote sensing image–voice retrieval. To capture more information, the inception dilated convolution module is developed to capture multiscale contextual information of remote sensing images and voices, which can provide richer information for hash codes' learning. Furthermore, the deep features' similarity term is proposed to make paired similar deep features as close as possible and paired dissimilar deep features as mutually far as possible. In addition, considering the quantization error between hash-like codes and hash codes, the quantization error term is designed to effectively reduce the quantization error. Experiments illustrate that the DIVR approach can achieve better retrieval performance than other state-of-the-art cross-modal retrieval approaches.

The contributions are summarized into three aspects.

1) A novel cross-modal remote sensing image–voice retrieval framework is proposed to utilize multiscale contextual information of remote sensing images and voices to solve the problem of insufficient utilization of data characteristics for cross-modal similarity learning. Moreover, the deep features' similarity term is designed to enhance cross-modal similarity. To the best of our knowledge, it is the first work to leverage hash codes to carry out cross-modal remote sensing image–voice retrieval.
2) Due to the fact that the quantization error between hash-like codes and hash codes is inevitable, the quantization error term is developed to drive hash-like codes to approximate hash codes, which can effectively reduce the quantization error.
3) Experimental results on three remote sensing image–voice data sets show that cross-modal similarity is significantly enhanced and the learned hash codes can achieve better performance than other state-of-the-art hashing approaches.

The remainder is given as follows. Section II reviews related retrieval approaches. Section III introduces the detailed

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procedure of the DIVR approach. Section IV presents the experimental results. Section V elaborates the conclusions of DIVR approach.

II. Prior Work

In this section, we mainly review the literature about two aspects: unimodal retrieval approaches and cross-modal retrieval approaches.

A. Unimodal Retrieval Approaches

Unimodal retrieval conducts the similarity search of the data in the same modality. Some early works of unimodal retrieval approaches in remote sensing include [20]–[23]. These methods leverage low-level feature descriptors to perform remote sensing image retrieval. For example, [20] leverages local invariant features to carry out remote sensing image retrieval. Aptoula [21] utilizes global morphological texture descriptors to perform content-based remote sensing image retrieval. However, these approaches cannot capture the complex semantic structure of remote sensing images. With the development of deep learning, many remote sensing retrieval approaches based on deep learning [5]–[7], [10] have been proposed to learn the complex semantic structure of remote sensing images. For example, [5] introduces hashing-based approximate nearest neighbor search for fast and accurate image search and retrieval in large remote sensing data archives. Peng and Peng [6] proposed a novel partial randomness hashing approaches, which can solve the large-scale image retrieval problems. Ye *et al.* [10] presented multiple feature hashing learning approaches in large-scale remote sensing image retrieval. Compared with unimodal retrieval approaches, cross-modal retrieval approaches are more challenging due to the heterogeneity of cross-modal data.

B. Cross-Modal Retrieval Approaches

Cross-modal retrieval mainly leverages the data of one modality as a query to retrieve the data of another modality. Cross-modal retrieval has attracted wide attentions in image and text [2], [24], [25], as well as in image and voice [13], [15], [26]–[30]. However, there are few previous studies on cross-modal retrieval in the remote sensing field. In this context, [31] proposed a new source-invariant deep hashing convolutional neural networks (CNNs), which can leverage dual-source remote sensing image data set to perform cross-source remote sensing image retrieval. Another attempt in cross-modal remote sensing retrieval is [12], which presents a novel deep image–voice framework to learn cross-modal similarity and performs cross-modal remote sensing image–voice retrieval. As far as we know, this is the only cross-modal image–voice approach in the remote sensing field.

However, existing cross-modal image–voice approach [12] in the remote sensing field cannot adequately consider the characteristics of remote sensing data and uses high-dimensional real-value features for retrieval. To solve these problems, the proposed DIVR approach not only designs inception dilated convolution module to capture multiscale contextual information of remote sensing images and voices but also utilizes the quantization error term and the deep features' similarity term to perform hash codes' learning simultaneously.

III. Proposed Method

In this section, we introduce the problem definition in Section III-A. Section III-B clarifies the details of the proposed architecture. The objective function of DIVR is presented in Section III-C.

A. Problem Definition

Given N pairwise units $\boldsymbol{\mathcal{P}} = \{I_j, V_j\}_{j=1}^{N}$ and N pairwise labels $\boldsymbol{\mathcal{S}} = \{s_j\}_{j=1}^{N}$, where $\boldsymbol{\mathcal{I}} = \{I_j\}_{j=1}^{N}$ represents the remote sensing image modality, and $\boldsymbol{\mathcal{V}} = \{V_j\}_{j=1}^{N}$ represents the corresponding voice modality. $s_j \in \{0, 1\}$, $s_j = 1$ represents that remote sensing image I_j and voice V_j contain similar concepts, $s_j = 0$ represents that I_j and V_j do not contain similar concepts. The goal of the proposed DIVR approach is to find two valid mapping function $\mathcal{H}_I$: $\boldsymbol{\mathcal{I}} = \{I_j\}_{j=1}^{N} \rightarrow \{-1, 1\}^{k\times N}$ and $\mathcal{H}_V$: $\boldsymbol{\mathcal{V}} = \{V_j\}_{j=1}^{N} \rightarrow \{-1, 1\}^{k\times N}$ for the two modalities while maintaining the similarity of similar remote sensing images and voices [32], where k is the length of the hash code. The similarity of paired remote sensing images and voices is assessed by leveraging the Hamming distance. Specially, if $S_j = 0$, the Hamming distance between hash codes $h_j^I = \mathcal{H}_I(I_j)$ and hash codes $h_j^V = \mathcal{H}_V(V_j)$ is large [33], and if $S_j = 1$, the Hamming distance between h_j^I and h_j^V is small.

B. Multimodal Architecture

The proposed deep cross-modal image–voice retrieval framework is shown in Fig. 2. The whole framework consists of the remote sensing image branch and the remote sensing voice branch.

1) Remote Sensing Image Branch: The configuration of the remote sensing image branch is shown in Fig. 2. The network architecture contains three 2-D convolutional layers with max-pooling and BN [34], [35] ($L_1 - L_3$), an inception dilated convolution module (L_4), and two fully connected layers (L_5 and L_6). These convolutional layers leverage 128, 128, and 256 3×3 filters with strides 1, respectively. The pooling layers exploit 2×2 windows. The inception dilated convolution module L_4 consists of one-dilated 3×3 convolution, two-dilated 3×3 convolution, and three-dilated 3×3 convolution. Note that the "one-dilated" specifies dilated convolution with dilated factor 1, the "two-dilated" specifies dilated convolution with dilated factor 2, and the "three-dilated" specifies dilated convolution with dilated factor 3. In 2-D, dilated convolution is given as

$$O[i] = \sum_{k} x[i + r * k]w[k] \tag{1}$$

where $x[i]$ is the input feature map, $O[i]$ is the output feature map, r represents the dilation rate, and w represents the filter. GoogLeNet naive inception module [36] contains three convolution with kernel size 1×1, 3×3, and 5×5, which can learn multiscale information. Inspired by the GoogLeNet

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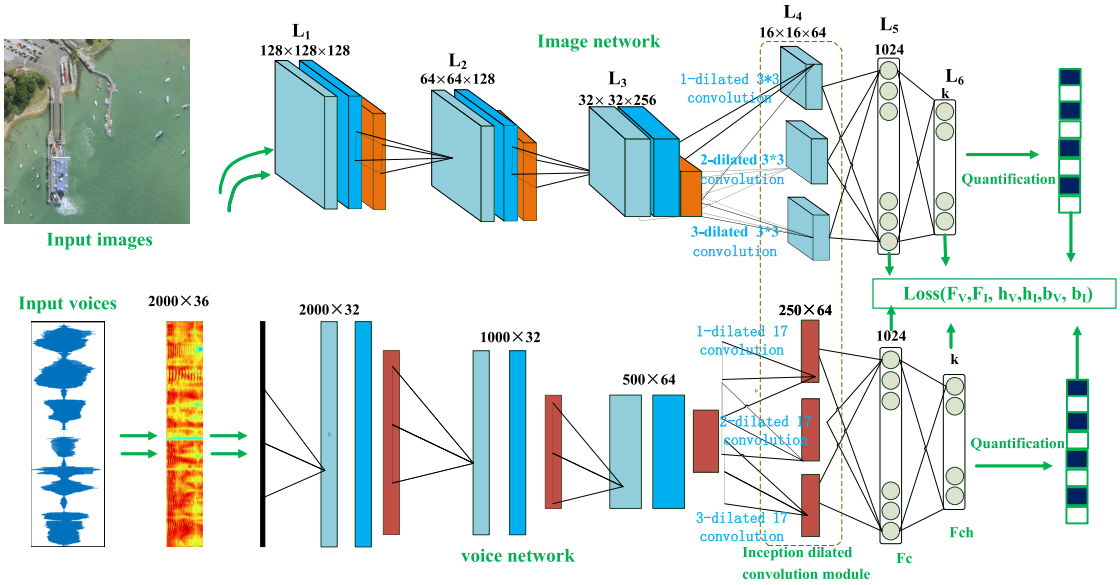


Fig. 2. Overview of the proposed DIVR framework. The whole framework contains the remote sensing image branch and the remote sensing voice branch. The remote sensing image branch contains three 2-D convolutional layers with max-pooling and BN, inception dilated convolution module, and two fully connected layers. For the remote sensing voice branch, the first convolution layer exploits filters with the width of one frame across the whole frequency axis, and then, the following layer of the remote sensing voice branch consists of three 1-D convolutions with BN and max-pooling, inception dilated convolution module, and two fully connected layers. The overall objective function consists of the semantic similarity term, the quantization error term, and the deep features' similarity term.

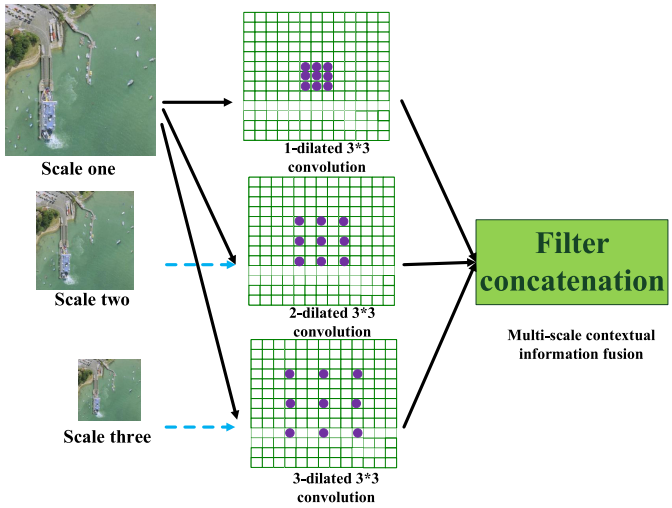


Fig. 3. Detailed operation process of the proposed inception dilated convolution module. It consists of one-dilated 3 × 3 convolution, two-dilated 3 × 3 convolution, and three-dilated 3 × 3 convolution. The black solid arrow represents that the same scale images of the previous layers are operated by three different dilated convolution, while the light blue dotted arrow represents that different scale image information can be learned by different dilated convolutions. In addition, the output of different dilated convolution operations is in the same size as the input. Furthermore, different scale contextual information can be easily aggregated by a concatenation operator.

naive inception module, we propose new inception dilated convolution module to capture multiscale contextual information of remote sensing image [37], as shown in Fig. 3. The black solid arrow represents that the same scale images of the previous layers are operated by three different dilated convolution, while the light blue dotted arrow represents different scale image information that can be learned by different dilated convolution. In addition, the output of different dilated convolution operations is the same size as the input. Furthermore, different scale contextual information can be easily aggregated by the concatenation operator. Thus, the inception dilated convolution module can capture multiscale contextual information.

The first fully connected layer L_5 exploits 1024 nodes, and the second fully connected layer L_6 is hash layer, which contains k units. The rectified linear unit (ReLU) [38], [39] is exploited as activation function for the layers L_{1-4}. The activation function of the layer L_5 is sigmoid function. The activation function of the layer L_6 is tanh function, which is an approximate quantization function of saturation nonlinearity and can generate k-bit hash-like codes q_j^I in the training stage and the testing stage. Then, hash codes h_j^I can be obtained by

$$h_j^I = \mathcal{H}_I(I_j) = \text{sign}(q_j^I) = \text{sign}(\Gamma(F_I(I_j), \theta_I)) \tag{2}$$

with

$$sign(x) = \begin{cases} 1, & x \geqslant 0 \\ -1, & x < 0 \end{cases} \tag{3}$$

where h_j^I denotes k-bit hash codes of remote sensing image I_j. q_j^I denotes k-bit hash-like codes of remote sensing image I_j. Γ denotes the tanh function. $F_I(I_j)$ denotes deep features, which is the output of the first fully connected layer in the remote sensing image branch. θ_I denotes the weight parameters of the hash layer in remote sensing image branch.

2) Remote Sensing Voice Branch: Following [40], mel-frequency cepstral coefficients (MFCCs) are adopted to represent remote sensing voice. It can be calculated by a-ms window size and b-ms shift. Moreover, the length of MFCC is forcibly converted to size c, truncated more than size c, and padded zero less than size c. The remote sensing voice branch is shown in Fig. 2. The first convolution layer of the remote

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sensing voice branch exploits filters with the width of one frame across the whole frequency axis. Then, the following layer of the remote sensing voice branch consists of three 1-D convolutions with BN [35] and max-pooling (L_1–L_3), inception dilated convolution module (L_4), and two fully connected layers (L_5 and L_6). These convolutional layers leverage 32, 32, and 64 filters. The corresponding respective widths are 11, 17, and 19, respectively. All pooling operations exploit two strides. The inception dilated convolution module consists of one-dilated 17 convolutions, two-dilated 17 convolutions, and three-dilated 17 convolutions, which can capture multiscale contextual information of remote sensing voice. In 1-D, dilated convolution is given as

$$g[i] = \sum_{l=1}^{L} f[i + r * l]h[l] \tag{4}$$

where $f[i]$ is the 1-D feature input, $g[i]$ is the 1-D feature output, $h[l]$ denotes the filter of length L, and r is the dilation rate.

The first fully connected layer L_5 contains 1024 nodes, and the second fully connected layer L_6 is hash layer, which exploits k units. The ReLU [38] is adopted as activation function for the layers L_{1-4}. The activation function of the layer L_5 is sigmoid function. The activation function of the hash layer L_6 is tanh function, which is an approximate quantization function of saturation nonlinearity and can generate k-bit hash-like codes. For remote sensing voice branch, hash codes can be obtained as

$$h_j^V = \mathcal{H}_V\left(V_j\right) = \operatorname{sign}\left(q_j^V\right) = \operatorname{sign}\left(\Gamma\left(F_V\left(V_j\right), \theta_V\right)\right) \tag{5}$$

where h_j^V denotes k-bit hash codes of remote sensing voice V_j. q_j^V denotes k-bit hash-like codes of remote sensing voice V_j. $F_V(V_j)$ denotes deep features, which is the output of the first fully connected layer in remote sensing voice branch. θ_V denotes the weight parameters of the hash layer in remote sensing voice branch.

C. Objective Function

Different from these cross-modal remote sensing retrieval approaches [12], [27], our proposed approach leverages hash codes to carry out cross-modal image–voice retrieval. The goal of the proposed DIVR approach is to find two valid mapping functions $\mathcal{H}_I$ and $\mathcal{H}_V$ for the two modalities while maintaining the similarity of similar remote sensing images and voices [32]. For this purpose, paired similar remote sensing image and voice are as close as possible, and paired dissimilar remote sensing image and voice are as mutually far as possible [41]. Inspired by the contrastive loss [34], the semantic similarity term can be defined as

$$\mathscr{L}_S = \sum_{j=1}^{N} \frac{1}{2}\left(1 - s_j\right)\max(m - H_{IV}, 0)^2 + \frac{1}{2}s_j H_{IV}^2 \tag{6}$$

where H_{IV} represents the Hamming distance for hash codes h_j^V of remote sensing image and hash codes h_j^I of remote sensing voice. $s_j \in \{0, 1\}$ represents the pairwise label. $\max(\cdot)$ represents the maximum function. m represents a margin threshold parameter. The semantic similarity term $\mathscr{L}_S$ can make paired similar remote sensing image and voice as close as possible and paired dissimilar remote sensing image and voice as mutually far as possible.

Directly optimizing (6) is infeasible since the calculation of Hamming distance makes the network difficult to train [34]. The binary constraints on hash codes require thresholding the network outputs and will make it intractable to train the network with a backpropagation algorithm. Inspired by [34], the novel relaxation strategy is proposed to utilize the Euclidean distance of hash-like codes to replace the Hamming distance of hash codes. Hash-like codes are approximate continuous values of hash codes without binary constraints, which can be trained with the backpropagation algorithm in the network. Then, (6) is redefined as

$$\mathscr{L}_S = \sum_{j=1}^{N} \frac{1}{2}\left(1 - s_j\right)\max(m - E_{IV}, 0)^2 + \frac{1}{2}s_j E_{IV}^2 \tag{7}$$

where E_{IV} represents the Euclidean distance for hash-like codes q_j^V of remote sensing image and hash-like codes q_j^I of remote sensing voice.

However, the proposed approach utilizes the relaxation strategy to learn hash-like codes and then converts hash-like codes to hash codes by quantization operations, which results in the quantization error. Motivated by ITQ [42], a quantization error term is proposed to reduce the quantization error. Considering the semantic similarity term and the quantization error term, the objective function can be defined as

$$\begin{aligned}\mathscr{L} &= L_S + \gamma L_Q \\ &= \sum_{j=1}^{N}\left(\frac{1}{2}\left(1 - s_j\right)\max(m - E_{IV}, 0)^2 + \frac{1}{2}s_j E_{IV}^2\right) \\ &\quad + \gamma\left(\sum_{j=1}^{N}\left\|q_j^I - h_j^I\right\|_2^2 + \sum_{j=1}^{N}\left\|q_j^V - h_j^V\right\|_2^2\right)\end{aligned} \tag{8}$$

where γ is the tradeoff parameter. $||\cdot||_2$ represents the l_2-norm of vector. The first term L_S is the semantic similarity term, which cam preserve the semantic similarity property of hash codes. The second term L_Q is the quantization error term, which can be effectively leveraged to reduce the quantization error.

Equation (8) utilizes l_2-norm to reduce the quantization error between hash-like codes and hash codes. Compared with l_2-norm, l_1-norm encourages sparsity in the network training [39], [43]. More hash codes are projected into 1 or -1, which can swimmingly produce compact hash codes. Then, (8) is redefined as

$$\begin{aligned}\mathscr{L} &= L_S + \gamma L_Q \\ &= \sum_{j=1}^{N}\left(\frac{1}{2}\left(1 - s_j\right)\max(m - E_{IV}, 0)^2 + \frac{1}{2}s_j E_{IV}^2\right) \\ &\quad + \gamma\left(\sum_{j=1}^{N}\left|\left|\left|q_j^I\right| - \left|h_j^I\right|\right|\right|_1^2 + \sum_{j=1}^{N}\left|\left|\left|q_j^V\right| - \left|h_j^V\right|\right|\right|_1^2\right)\end{aligned} \tag{9}$$

where $||\cdot||_1$ represents the l_1-norm of vector. $|\cdot|$ represents the absolute value operation.

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Nevertheless, the regularizer term is hard to compute the derivative on the proposed network framework. Motivated by [44], the smooth surrogate of the absolute function $|x| = \log\cosh x$ is utilized in the proposed network framework. The absolute function $|x|$ is not differentiable at zero because its slope abruptly changes from -1 to $+1$. Thus, it is useful to take a smoother function $\log\cosh x$ to replace the absolute function $|x|$ in the optimization algorithms. Equation (9) is redefined as

$$\begin{aligned}\mathscr{L} &= L_S + \gamma L_Q \\ &= \sum_{j=1}^{N}\left(\frac{1}{2}(1-s_j)\max\left(m-E_{ij},0\right)^2 + \frac{1}{2}s_j E_{ij}^2\right) \\ &\quad + \gamma\left(\sum_{j=1}^{N}\sum_{k=1}^{K}\left(\log\cosh\left(\left|q_{j(k)}^{I}\right| - \left|h_{j(k)}^{I}\right|\right)\right)^2\right. \\ &\quad \left. + \sum_{j=1}^{N}\sum_{k=1}^{K}\left(\log\cosh\left(\left|h_{j(k)}^{V}\right| - \left|q_{j(k)}^{V}\right|\right)\right)^2\right) \quad (10)\end{aligned}$$

where $h_{j(k)}^{I}$ and $q_{j(k)}^{I}$ represent the kth position of hash codes h_j^I and the kth position of hash-like codes q_i^I for remote sensing image I_j, respectively. K represents the length of hash codes. $h_{j(k)}^{V}$ and $q_{j(k)}^{V}$ represent the kth position of hash codes h_j^V and the kth position of hash-like codes q_i^V for remote sensing voice V_j, respectively.

Although the abovementioned objective function (9) considers the semantic similarity property of hash codes, deep features' similarity is not fully exploited to produce hash codes because deep features' similarity can enhance the similarity property of hash codes. In order to enhance deep features' similarity, we add the deep features' similarity term in the objective function. The overall objective function $\mathscr{L}$ of the proposed DIVR approach can be redefined as

$$\begin{aligned}\mathscr{L} &= L_S + \gamma L_Q + \lambda L_D \\ &= \sum_{j=1}^{N}\left(\frac{1}{2}(1-s_j)\max(m-E_{IV},0)^2 + \frac{1}{2}s_j E_{IV}^2\right) \\ &\quad + \gamma\left(\sum_{i=1}^{N}\sum_{k=1}^{K}\left(\log\cosh\left(|b_i^{(k)}| - |h_i^{(k)}|\right)\right)^2\right. \\ &\quad \left. + \sum_{j=1}^{N}\sum_{k=1}^{K}\left(\log\cosh\left(|h_j^{(k)}| - |b_j^{(k)}|\right)\right)^2\right) \\ &\quad + \lambda\sum_{j=1}^{N}\left(\frac{1}{2}(1-s_j)\max(m-D_{IV},0)^2 + \frac{1}{2}s_j D_{IV}^2\right) \quad (11)\end{aligned}$$

where γ and λ represent the tradeoff parameters. D_{IV} represents the distance for deep features $F_I(I_j)$ of remote sensing image and deep features $F_V(V_j)$ of remote sensing voice. The overall objective function (11) is convex and can be optimized by Adam [45]. The optimization algorithm for learning DIVR can be listed in Algorithm 1. Furthermore, the objective function (11) consists of the semantic similarity term L_S, the quantization error term L_Q, and the deep features' similarity term L_D. The semantic similarity term L_S can make paired similar hash codes of remote sensing image and voice as close as possible and paired dissimilar hash codes of remote sensing image and voice as mutually far as possible. The quantization error term L_Q can be effectively leveraged to reduce the quantization error. The deep features' similarity term L_D can make paired similar deep features as close as possible and paired dissimilar deep features as mutually far as possible.

Algorithm 1 Optimization Algorithm for Learning DIVR

Input:

N pairwise units $\mathcal{P} = \{I_j, V_j\}_{j=1}^{N}$ and N pairwise labels $\mathcal{S} = \{s_j\}_{j=1}^{N}$

Output:

The parameters W of the proposed network framework;

Initialization:

Initialized W by using glorot_uniform distribution.

Repeat:

1: Pick pairwise units I_j, V_j and pairwise label s_j.
2: Calculate MFCC to represent remote sensing voice V_j.
3: Calculate deep features $F_I(I_j)$, $F_V(V_j)$ and hash-like codes q_j^V,q_j^I by forward propagation;
4: Calculate hash codes h_j^I and h_j^V by using Eq. (2) and Eq. (5) ;
5: Leverage $F_I(I_j)$, $F_V(V_j)$, q_j^V,q_j^I, h_j^I and h_j^V to calculate $\mathscr{L}$ according to Eq. (11);
6: Update W by utilizing Adam;

Until:

A fixed number of iterations or a stopping criteria is satisfied

Return: W

IV. Experiments

In this section, we first elaborate three remote sensing image–voice data sets and evaluation protocols in Section IV-A. Section IV-B introduces implementation details of the experiment. Section IV-C carries on the parameter analysis. Section IV-D analyzes different factors of the proposed DIVR approach. Section IV-E compares the proposed DIVR approach with other cross-modal retrieval algorithms.

A. Data Set and Evaluation Protocols

To verify the effectiveness of the proposed DIVR approach, we leverage three remote sensing image–voice data sets to compare with other cross-modal retrieval algorithms.

1) UC Merced data set (UCM) [46] consists of 2100 remote sensing image–voice pairs. Note that the original data set consists of 2100 remote sensing images, which can be divided into 21 classes, and each class contains 100 images. Each remote sensing image has one voice. Following [12], we construct 4200 positive and negative remote sensing image–voice pairs. In the training stage, 80% remote sensing image–voice pairs are randomly chosen as the training set, and other 20% remote sensing image–voice pairs are chosen as the testing set. In the testing stage, we exploit testing remote sensing images

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(respectively, voices) as the query set and testing remote sensing voices (respectively, remote sensing images) as the gallery set.

2) Sydney data set [46] contains 613 remote sensing image–voice pairs. Note that the original data set consists of 613 remote sensing images of seven classes; each remote sensing image has one voice. Following [12], we construct 1226 positive and negative remote sensing image–voice pairs and select 80% image–voice pairs as the training set. The rest pairs are chosen as the testing set. In the testing stage, we exploit testing remote sensing images (respectively, voices) as the query set and testing remote sensing voices (respectively, remote sensing images) as the gallery set.
3) Remote sensing image captioning data set (RSICD) [47] contains 10 921 remote sensing image–voice pairs. Note that the original data set consists of 10 921 remote sensing images of 30 classes; each remote sensing image has one voice. Following [12], we construct 21 842 positive and negative remote sensing image–voice pairs. We randomly select 20% remote sensing image–voice pairs as the testing set and other 80% remote sensing image–voice pairs as the training set. In the testing stage, we exploit testing remote sensing images (respectively, voices) as the query set and testing remote sensing voices (respectively, remote sensing images) as the gallery set.

In order to accomplish the cross-modal task of voice and image, the standard American pronunciation is used to generate spoken captions. To better meet the needs of voice diversity, different speakers participate in the collection of the data set. The ground-truth neighbors are defined as image–voice pairs that contain identical concepts. The experimental results are evaluated by mean average precision (MAP) and the precision in the top m of the retrieval list (precision@m). MAP is calculated as the mean of the average precision scores for each query. The precision@m is calculated as the average precision of top m returned neighbors of the ranked list. If the values of these metrics are higher, the performance of the approach is better.

To assess the performance of the proposed DIVR approach, we compare our DIVR approach with scale-invariant feature transform+M (SIFT+M), deep basic language processing (DBLP) [27], CNN+spectrum (CNN+SPEC) [29], deep visual audio network (DVAN) [12], cross-modal image retrieval network (CMIR-NET) [11], DIVR-D, DIVR-Q and DIVR-I. DIVR-D, DIVR-Q, and DIVR-I that are the variants of the proposed DIVR approach. The SIFT+M approach first extracts SIFT features of images and MFCC features of sounds and then maps their features in a common space by using the pairwise loss. The DBLP approach investigates novel neural network architectures for the purpose of learning high-level semantic concepts across both audio and visual modalities. The CNN+SPEC approaches a novel multimodal long short-term memory (LSTM) architecture, which seamlessly unifies both visual and auditory modalities from the raw input. The DVAN approach presents a novel deep approach

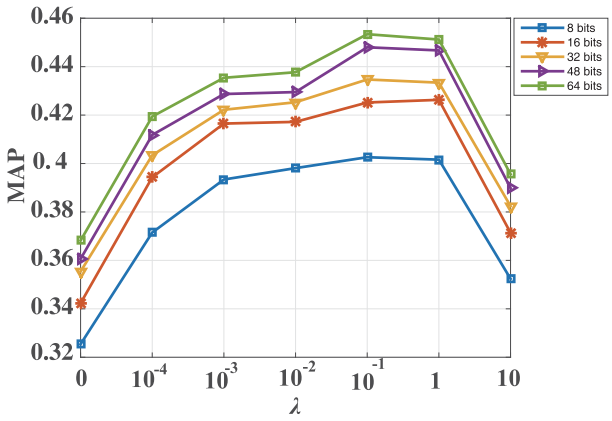


Fig. 4. MAP varies with different values of the parameter λ for different hash bits on the UCM data set by utilizing voices to retrieve remote sensing images.

image–voice frameworks to learn cross-modal similarity and performs cross-modal remote sensing image–voice retrieval. The CMIR-NET approach proposes a novel deep neural network-based architecture, which is considered to learn a discriminative shared feature space for all the input modalities. These approaches, i.e., DBLP [27], CNN+SPEC [29], DVAN [12], and CMIR-NET [11], were carried out by these authors' articles. The proposed DIVR approach exploits 64 bit hash codes to compare with SIFT+M, DBLP [27], CNN+SPEC [29], DVAN [12], and CMIR-NET [11].

B. Implementation Detail

The proposed DIVR approach leverages the open-source KERAS[1] library [48] to carry out the experiments on a workstation with GeForce GTX Titan X GPU, Inter Core i7−5930K 3.50-GHZ CPU, and 64-GB RAM. The overall objective function (11) is optimized by Adam [45] with the learning rate 10^{-3}. The batch size is set to 64. To generate {8, 16, 24, 36, 48, 64}-bit hash codes, k is set from 8 to 64, respectively. The initial weights of the proposed network framework utilize glorot_uniform distribution. The parameter γ is set as 1, and the parameter λ is set as 0.1. The parameter a is set as 16, the parameter b is set as 5, and the parameter c is set as 2000. For the UCM data set, the proposed network is trained 1800 epochs or stops training until the loss does not diminish. For the Sydney data set, the proposed network is trained 1000 epochs or stops training until the loss does not diminish. For RSICD data set, the proposed network is trained 3000 epochs or stops training until the loss does not diminish [49].

C. Parameter Analysis

To perform parameters' analysis for the proposed DIVR approach, we implement experiments about the two parameters γ and λ of (11) on the UCM data set. We first fix the parameter $\gamma = 1$. MAP varies with different values of the parameter λ for different hash bits on the UCM data set by utilizing voices to retrieve remote sensing images, as shown in Fig. 4. It can be shown from Fig. 4 that the proposed method can obtain the best MAP when the parameter λ is set as 0.1. Then, we fix the

[1]https://github.com/fchollet/keras

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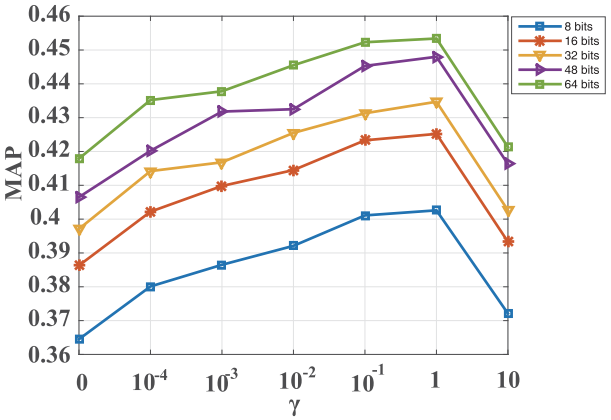


Fig. 5. MAP varies with different values of the parameter γ for different hash bits on the UCM data set by utilizing voices to retrieve remote sensing images.

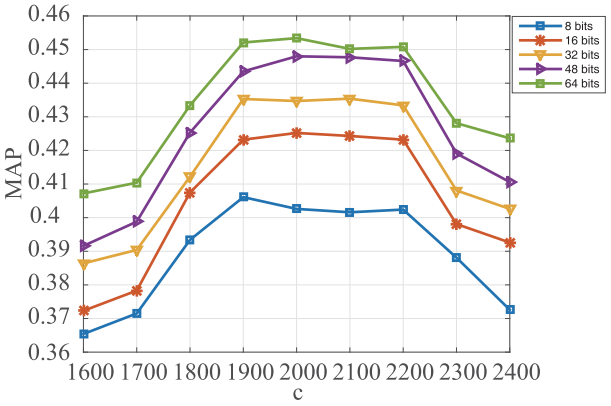


Fig. 6. MAP varies with different values of the parameter c for different hash bits on the UCM data set by utilizing voices to retrieve remote sensing images.

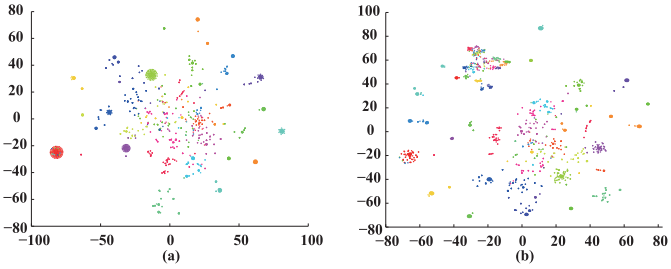


Fig. 7. t-SNE visualizations of deep features of the DIVR-P approach for (a) remote sensing image on UCM data set and (b) remote sensing voice on UCM data set.

parameter $\lambda = 0.1$, and MAP varies with different values of the parameter γ for different hash bits on the UCM data set by utilizing voices to retrieve remote sensing images, as shown in Fig. 5. It can be seen from Fig. 5 that the proposed DIVR approach can obtain the best MAP when the parameter λ is set as γ. Hence, the parameters λ and γ in the proposed DIVR approach are set as 0.1 and 1, respectively.

MFCCs are adopted to represent remote sensing voice. It can be calculated by a-ms window size and b-ms shift. Moreover, the length of MFCC is forcibly converted to size c, truncated more than size c, and padded zero less than

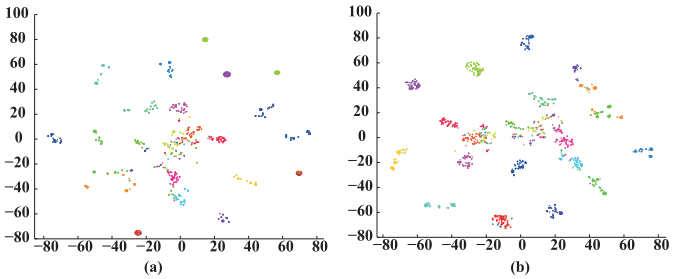


Fig. 8. t-SNE visualizations of deep features of the DIVR approach for (a) remote sensing image on UCM data set and (**b**) remote sensing voice on UCM data set.

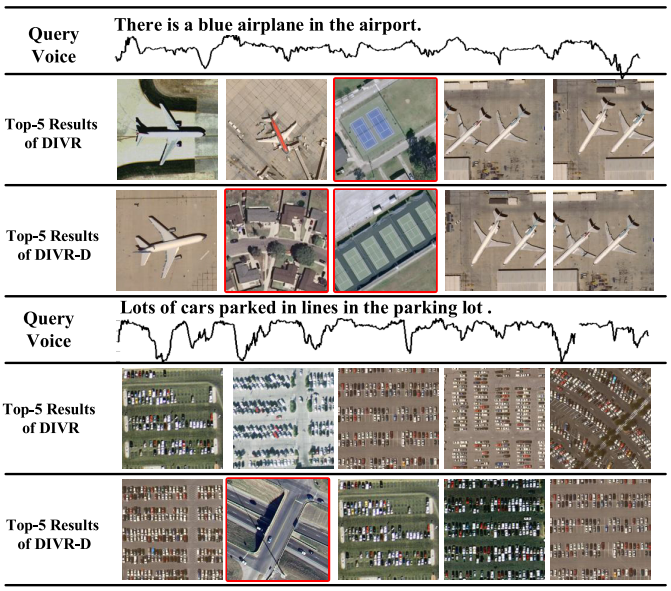


Fig. 9. Comparative top-five retrieval results between DIVR and DIVR-D on UCM data set by utilizing voices to retrieve remote sensing images. The wrong retrieval results are marked with red font.

size c. According to these studies [12], [40], [50], a and b are commonly used parameters for calculating MFCC. In the process of extracting voice features, the parameters a and b are commonly set to 5 and 16, respectively. However, the parameter c requires optimal selection for the proposed DIVR approach. Following [12], [40], [50], the parameter c is set from 1600 to 2400. MAP varies with different values of the parameter c for different hash bits on the UCM data set by utilizing voices to retrieve remote sensing images, as shown in Fig. 6. It can be seen from Fig. 6 that the proposed DIVR approach can obtain the best MAP when the parameter c is set as 2000. Hence, the parameters c in the proposed DIVR approach are set as 2000.

D. Evaluation of Different Factors

To evaluate the effectiveness of exploiting the inception dilated convolution module, the quantization error term, and the deep features' similarity term in the DIVR approach, we carry out the experiment in the following four aspects. First, we exploit the proposed approach without exploiting the inception dilated convolution module (i.e., DIVR-I).

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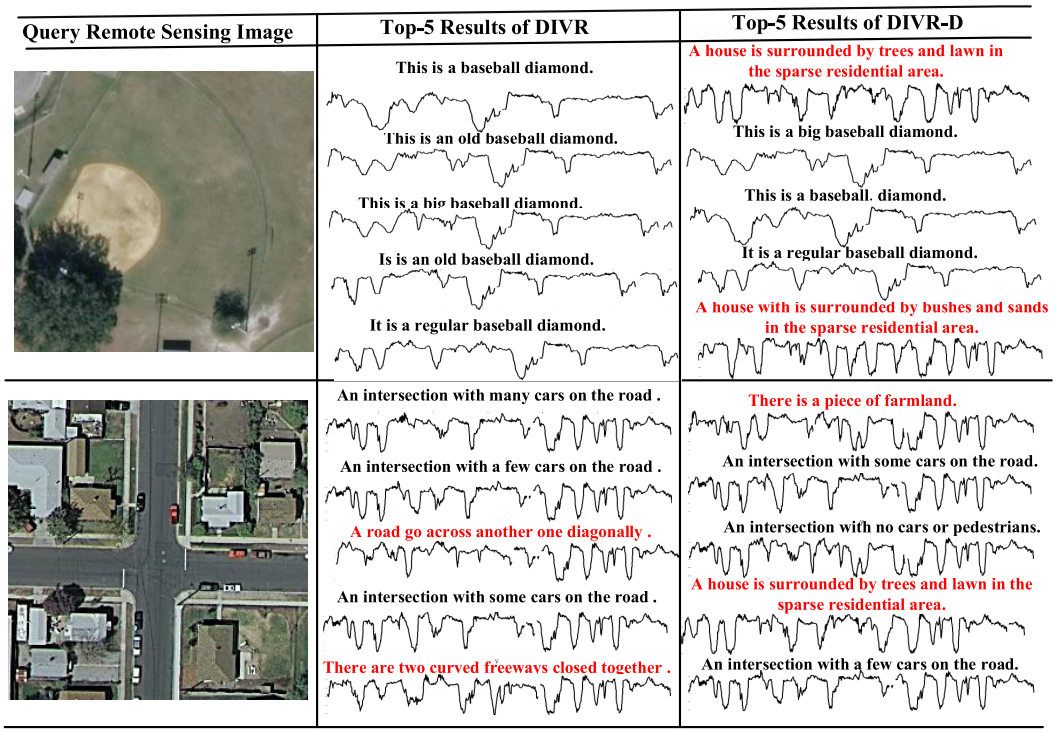


Fig. 10. Comparative top-five retrieval results between DIVR and DIVR-D on UCM data set by utilizing remote sensing images to retrieve voices. The wrong retrieval results are marked with red font.

TABLE I

COMPARISON OF DIVR-P, DIVR-D, DIVR-Q, DIVR-I, AND DIVR ON UCM DATA SET WITH DIFFERENT HASH BITS

	Constraint	16 bits	32 bits	48 bits	64 bits
V→ I	DIVR-P	30.31	31.15	31.66	32.12
	DIVR-D	34.23	35.54	36.06	36.82
	DIVR-Q	38.63	39.71	40.65	41.79
	DIVR-I	39.93	41.15	41.74	43.18
	DIVR	42.52	43.47	44.80	45.34
I→ V	DIVR-P	32.96	34.27	34.98	35.83
	DIVR-D	40.34	41.57	42.14	42.58
	DIVR-Q	43.79	45.04	45.73	46.34
	DIVR-I	44.85	45.86	46.94	47.72
	DIVR	48.53	49.14	49.81	50.94

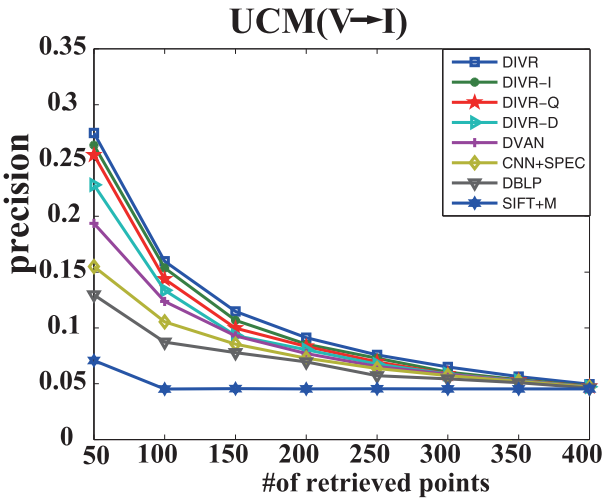


Fig. 11. Precision curves with different returned images by utilizing voices to retrieve remote sensing images on UCM data set.

Second, we use the proposed approach without exploiting the quantization error term L_Q (i.e., DIVR-Q). Third, we utilize the proposed approach without exploiting the deep features' similarity term L_D (i.e., DIVR-D). Finally, we leverage the proposed approach without exploiting the inception dilated convolution module, the quantization error term, and the deep features' similarity term (i.e., DIVR-P). Table I shows contrastive results of DIVR-P, DIVR-D, DIVR-Q, DIVR-I, and DIVR on the UCM data set with different hash bits. Here, "V→I" denotes the case where the query sets are remote sensing voices and the gallery sets are remote sensing images. "I→V" denotes the case where the query sets are remote sensing images and the gallery sets are remote sensing voice. From Table I, we can see clearly that the proposed DIVR approach can outperform DIVR-P, DIVR-D, DIVR-Q, and DIVR-I on the MAP with different hash bits. For V→I, the proposed DIVR approach can improve the MAP result with 48 bits from DIVR-P (31.66%), DIVR-D (36.06%), DIVR-Q (40.65%), and DIVR-I (41.74%) to 44.80%. For I→V, the proposed DIVR approach can improve the MAP result with 48 bits from DIVR-P (34.98%), DIVR-D (42.14%), DIVR-Q (45.73%), and DIVR-I (46.94%) to 49.81%. This is because the proposed DIVR approach exploits the inception dilated convolution module, the quantization error term, and the deep features' similarity term to achieve better search performance.

Moreover, Fig. 7(a) shows t-SNE visualizations of deep features of the DIVR-P approach for remote sensing image

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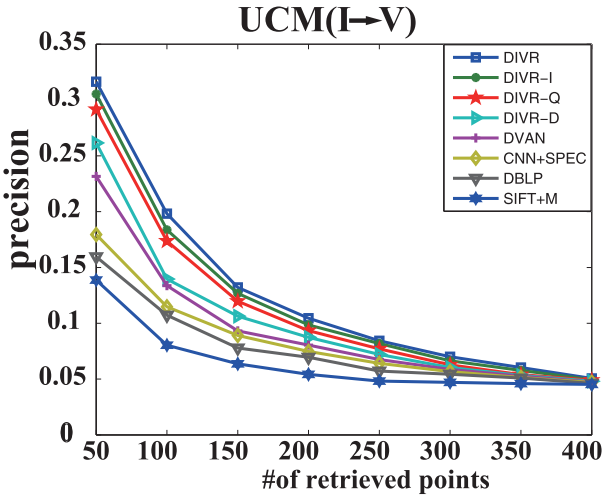


Fig. 12. Precision curves with different returned images by utilizing remote sensing images to retrieve voices on UCM data set.

TABLE II

COMPARISON OF EXPERIMENTAL RESULTS WITH UTILIZING VOICES TO RETRIEVE REMOTE SENSING IMAGES IN OUR PROPOSED DIVR APPROACH AND OTHER APPROACHES ON UCM DATA SET

Task	Method	MAP	P@1	P@5	P@10
	SIFT+M	6.66	3.58	4.41	4.68
	DBLP [27]	19.33	17.12	17.62	16.31
	CNN+SPEC [29]	21.79	19.42	19.86	19.23
	DVAN [12]	32.28	32.37	33.91	34.34
V→ I	DIVR-D	36.82	42.76	39.48	35.94
	CMIR-NET[11]	40.37	46.74	43.75	39.62
	DIVR-Q	41.79	47.84	44.12	40.23
	DIVR-I	43.18	50.06	46.68	42.64
	DIVR	45.34	52.23	48.76	44.98

on the UCM data set. Fig. 7(b) shows t-SNE visualizations of deep features of the DIVR-P approach for remote sensing voice on UCM data set. Fig. 8(a) shows t-SNE visualizations of deep features of the DIVR approach for remote sensing image on UCM data set. Fig. 8(b) shows t-SNE visualizations of deep features of the DIVR approach remote sensing voice on UCM data set. It is apparent from Figs. 7 and 8 that the DIVR approach can learn more effective deep features than DIVR-P. In addition, we estimate comparison between DIVR without exploiting deep features' similarity and DIVR on the UCM data set in Figs. 9 and 10, where top-five retrieval results intuitively demonstrate the effectiveness of deep features' similarity.

E. Method Comparison

1) Results on UCM: Table II shows the compared results between the proposed DIVR approach and other comparative approaches on UCM data set by utilizing voices to retrieve remote sensing images. Table III shows the compared results between the proposed DIVR approach and other comparative approaches on the UCM data set by utilizing remote sensing images to retrieve voices. Fig. 11 shows precision curves

TABLE III

COMPARISON OF EXPERIMENTAL RESULTS WITH UTILIZING REMOTE SENSING IMAGES TO RETRIEVE VOICES IN OUR PROPOSED DIVR APPROACH AND OTHER APPROACHES ON UCM DATA SET

Task	Method	MAP	P@1	P@5	P@10
	SIFT+M	8.55	4.56	4.65	4.56
	DBLP [27]	25.48	24.18	23.87	23.24
	CNN+SPEC [29]	26.25	29.5	25.52	23.65
	DVAN [12]	36.79	32.37	33.29	33.74
I→ V	DIVR-D	42.58	50.46	46.34	41.95
	CMIR-NET[11]	45.82	52.92	49.74	43.38
	DIVR-Q	46.34	54.34	50.34	45.09
	DIVR-I	47.72	56.74	51.83	46.34
	DIVR	50.94	59.34	54.17	50.12

with different returned images by utilizing voices to retrieve remote sensing images on the UCM data set. Fig. 12 shows precision curves with different returned images by utilizing remote sensing images to retrieve voices on the UCM data set. We can clearly observe the following.

1) Although those compared approaches have achieved good performance, the proposed DIVR approach can acquire the highest precision value within MAP, the highest precision value within top one returned instance, the highest precision value within the top-five returned instance, and the highest precision value within the top-ten returned instance.
2) From these two figures, the proposed DIVR approach achieves better performance than those compared approaches at all retrieved neighbors for utilizing voices to retrieve remote sensing images or utilizing remote sensing images to retrieve voices.
3) For I→V, the proposed DIVR approach can improve the MAP result from SIFT+M (8.55%), DBLP (25.48%), CNN+SPEC (26.25%), DVAN (36.79%), DIVR-D (42.58%), CMIR-NET (45.82%), DIVR-Q (46.43%), and DIVR-I (47.72%) to 50.94%. Furthermore, for V→I, the proposed DIVR approach can improve the average MAP result from SIFT+M (6.66%), DBLP (19.33%), CNN+SPEC (21.79%), DVAN (32.28%), DIVR-D (36.82%), CMIR-NET (40.37%), DIVR-Q (41.79%), and DIVR-I (43.18%) to 45.34%. This is because the proposed DIVR approach not only utilizes the inception dilated convolution module to capture multiscale contextual information of remote sensing image and voice but also leverages the quantization error term and the deep features' similarity term to enhance hash codes' learning.

2) Results on Sydney: Table IV shows the compared results between the proposed DIVR approach and other comparative approaches on Sydney data set by utilizing voices to retrieve remote sensing images. Table V shows the compared results between the proposed DIVR approach and other comparative approaches on the Sydney data set by utilizing remote sensing images to retrieve voices. Precision curves with different returned images by utilizing voices to retrieve

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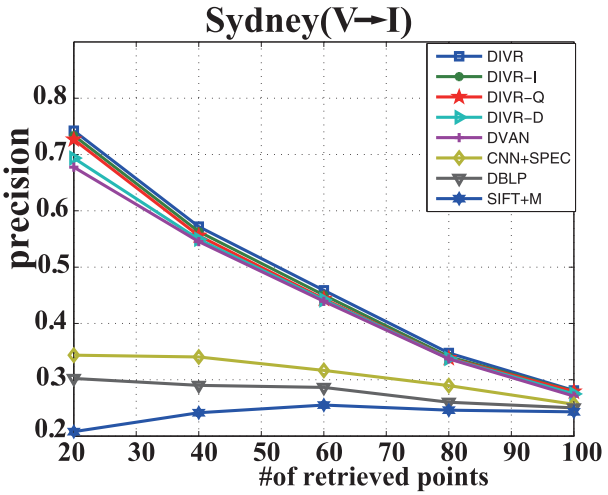


Fig. 13. Precision curves with different returned images by utilizing voices to retrieve remote sensing images on Sydney data set.

TABLE IV

COMPARISON OF EXPERIMENTAL RESULTS WITH UTILIZING VOICES TO RETRIEVE REMOTE SENSING IMAGES IN OUR PROPOSED DIVR APPROACH AND OTHER APPROACHES ON SYDNEY DATA SET

Task	Method	MAP	P@1	P@5	P@10
	SIFT+M	26.50	34.48	24.48	23.28
	DBLP [27]	34.87	21.63	26.78	30.94
	CNN+SPEC [29]	35.72	17.24	27.76	31.21
	DVAN [12]	63.88	67.24	63.34	67.07
V→ I	DIVR-D	69.65	74.37	71.47	69.42
	CMIR-NET[11]	71.28	76.69	74.52	71.60
	DIVR-Q	72.70	77.12	75.91	72.84
	DIVR-I	73.87	78.36	76.46	73.85
	DIVR	75.97	80.44	78.05	76.27

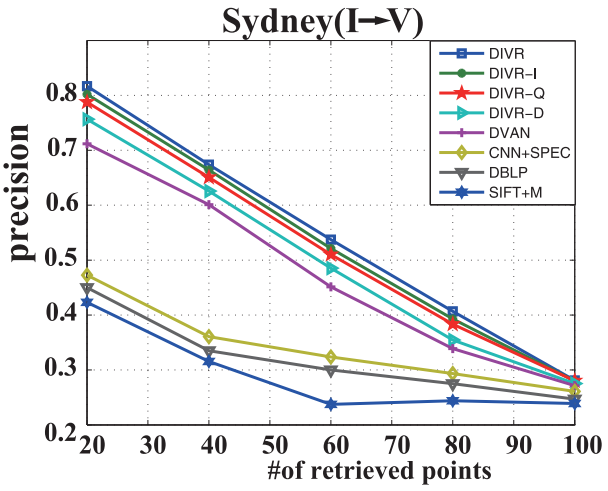


Fig. 14. Precision curves with different returned images by utilizing remote sensing images to retrieve voices on Sydney data set.

remote sensing images and precision curves with different returned images by utilizing remote sensing images to retrieve voices are shown in Figs. 13 and 14, respectively. Similar

TABLE V

COMPARISON OF EXPERIMENTAL RESULTS WITH UTILIZING REMOTE SENSING IMAGES TO RETRIEVE VOICES IN OUR PROPOSED DIVR APPROACH AND OTHER APPROACHES ON SYDNEY DATA SET

Task	Method	MAP	P@1	P@5	P@10
	SIFT+M	31.67	11.21	35.00	37.59
	DBLP [27]	44.38	56.51	52.65	49.68
	CNN+SPEC [29]	46.67	58.62	55.00	51.64
	DVAN [12]	71.77	75.86	73.62	72.93
I→ V	DIVR-D	75.90	82.72	80.37	78.68
	CMIR-NET[11]	78.44	84.68	82.54	81.04
	DIVR-Q	79.20	86.38	84.44	82.08
	DIVR-I	80.22	87.30	85.63	83.46
	DIVR	81.35	88.26	86.35	84.47

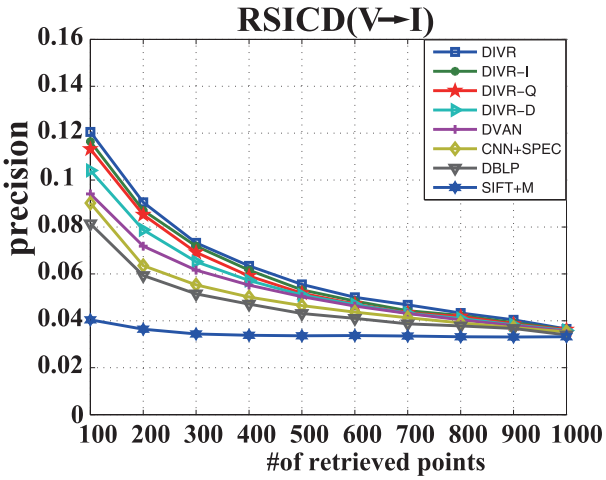


Fig. 15. Precision curves with different returned images by utilizing voices to retrieve remote sensing images on RSICD data set.

retrieval results are explicitly observed on the UCM data set. For example, for I→V, the proposed DIVR approach can improve the MAP result from SIFT+M (31.67%), DBLP (44.38%), CNN+SPEC (46.67%), DVAN (71.77%), DIVR-D (75.90%), CMIR-NET (78.44%), DIVR-Q (79.20%), and DIVR-I (80.22%) to 81.35%. Furthermore, for V→I, the proposed DIVR approach can improve the MAP result from SIFT+M (26.5%), DBLP (34.87%), CNN+SPEC (35.72%), DVAN (63.88%), DIVR-D (69.65%), CMIR-NET (71.28%), DIVR-Q (72.70%), and DIVR-I (73.87%) to 75.97%. The proposed DIVR approach can achieve the best results on all the evaluation metrics, which demonstrates the benefit of learning hash codes by exploiting the inception dilated convolution module, the quantization error term, and the deep features' similarity term simultaneously.

3) Results on RSICD Data Set: RSICD image–voice data set contains 54 605 image–voice pairs, which is more challenging than the UCM data set and RSICD data set. Tables VI and VII show the compared results between the proposed DIVR approach and other comparative approaches on the RSICD data set. Fig. 15 shows precision curves with different returned images utilizing voices to retrieve remote sensing images on the RSICD data set. Fig. 16 shows precision curves with

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TABLE VI

COMPARISON OF EXPERIMENTAL RESULTS IN OUR PROPOSED DIVR APPROACH AND OTHER APPROACHES ON RSICD DATA SET

Task	Method	MAP	P@1	P@5	P@10
V→ I	SIFT+M	4.85	3.66	3.60	3.54
	DBLP[27]	8.14	6.21	6.08	6.76
	CNN+SPEC [29]	9.96	7.13	7.00	7.44
	DVAN[12]	15.71	16.18	15.10	14.76
	DIVR-D	16.80	17.74	16.58	15.36
	CMIR-NET[11]	17.25	17.94	16.83	16.05
	DIVR-Q	17.43	18.29	17.18	16.53
	DIVR-I	17.96	18.82	17.83	17.16
	DIVR	18.58	19.76	18.31	17.59

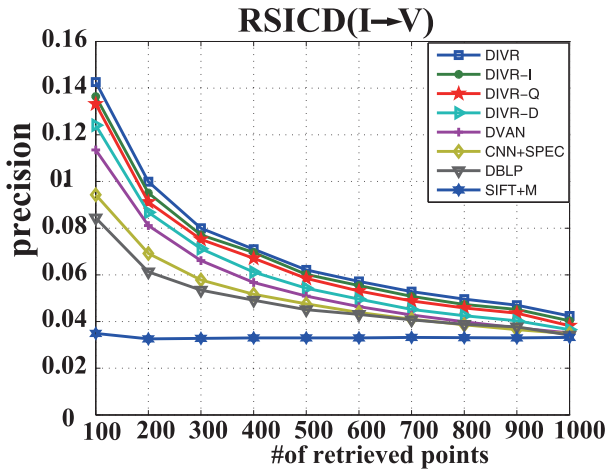


Fig. 16. Precision curves with different returned images by utilizing remote sensing images to retrieve voices on RSICD data set.

TABLE VII

COMPARISON OF EXPERIMENTAL RESULTS IN OUR PROPOSED DIVR APPROACH AND OTHER APPROACHES ON RSICD DATA SET

Task	Method	MAP	P@1	P@5	P@10
I→ V	SIFT+M	5.04	6.22	5.34	4.50
	DBLP[27]	12.70	15.32	15.21	14.22
	CNN+SPEC[29]	13.24	16.82	16.62	15.69
	DVAN[12]	16.29	22.49	22.56	21.7
	DIVR-D	17.56	23.77	23.28	22.67
	CMIR-NET[11]	17.78	24.11	23.52	22.54
	DIVR-Q	18.17	24.24	23.61	22.58
	DIVR-I	18.87	24.91	24.34	23.66
	DIVR	19.62	25.43	24.84	24.20

different returned images utilizing remote sensing images to retrieve voices on the RSICD data set. The proposed VVCMH approach can achieve best P@1, P@5, P@10, and MAP results, which is favorably against other state-of-the-art cross-modal retrieval approaches: SIFT+M, DBLP, CNN+SPEC, DVAN, and CMIR-NET. This further demonstrates the effectiveness of the proposed method.

V. CONCLUSION

In this article, we have proposed a novel DIVR approach, which utilizes the similarity of deep features and multiscale contextual information to guide low-dimensional binary representations learning in cross-modal remote sensing retrieval. Especially, compared with high-dimensional feature representations, low-dimensional binary features can save storage and improve retrieval speed. First, we propose novel inception dilated convolution module, which can capture multiscale contextual information of remote sensing images and voices. Second, the deep features' similarity term is designed to make paired similar deep features as close as possible and paired dissimilar deep features as mutually far as possible. Third, the quantization error term is proposed to reduce the quantization error between hash-like codes and hash codes. Finally, extensive experimental results on UCM, Sydney, and RSICD data sets show that the proposed DIVR approach can achieve better performance than other cross-modal retrieval approaches. Future work can be divided into two main aspects. First, we will focus on how to perform the panchromatic to multispectral or LiDAR to multispectral cross-modal image retrieval. Second, we plan to exploit the proposed DIVR approach in other applications, such as disaster relief and electronic voice-digital map to demonstrate its extensive effectiveness.

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